



RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

Bachelor of Science in Information Technology
First Year - Semester I Examination – February 2024

ICT 1201 –FUNDAMENTALS OF COMPUTER SYSTEMS

Time: Two (02) hours.

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- There are four (04) questions in four (04) pages.
 - Answer **ALL** questions.
 - For **Question 1**, write the question number and the correct option in the answer booklet provided.
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1. a) 1. What is the primary function of a computer?
 - a. Store data
 - b. Perform calculations
 - c. Entertain people
 - d. Communicate with other computers
2. What technology significantly reduced the size and energy consumption of second-generation computers?
 - a. Transistors
 - b. Vacuum tubes
 - c. Integrated Circuits (IC)
 - d. Magnetic drums
3. What term is used for finding and correcting errors in a computer program or electronic hardware?
 - a. Networking
 - b. Artificial intelligence
 - c. Punch card processing
 - d. Debugging

4. What is the main characteristic of digital computers?
 - a. Operate with continuously variable physical quantities
 - b. Process data into 0s and 1s
 - c. Measure analog quantities
 - d. Calculate differential equations
5. is an input/output device functions as a window into a mainframe computer.
 - a. Dumb terminal
 - b. Intelligent terminal
 - c. Mainframe terminal
 - d. Analog terminal
6. What is the largest decimal integer you can store in one byte?
 - a. 128
 - b. 256
 - c. 64
 - d. 255
7. What type of translation is performed by a compiler?
 - a. Line-by-line translation
 - b. On-the-fly translation
 - c. Entire program code translation
 - d. Bytecode translation
8. In the context of programming, what is syntax?
 - a. The meaning of the language instructions
 - b. The rules and guidelines for constructing sentences or instructions
 - c. A collection of terms valid for the language
 - d. The vocabulary used in the language
9. What is the main purpose of comments in a program?
 - a. Enhancing program aesthetics
 - b. Communicating with the compiler
 - c. Providing information to programmers and making the code more readable
 - d. Enhancing program execution speed
10. What is the purpose of the "return" statement in a function?
 - a. To end the function's execution
 - b. To return a value from the function
 - c. To print output to the console
 - d. To declare a variable

(2x10 marks)

b) Fill in the blanks with suitable word/ word phrases.

1. _____ are a key function that distinguishes Analytical Engine from rest of the calculators.
2. Based on the function, computers can be classified as _____ and _____ computers.
3. A bus that connects major components (CPU, Memory, I/O) is called a _____.
4. _____ is a software application that provides comprehensive facilities to computer programmers for Software development.
5. A software suit is a group of different but interrelated software programs that are combined and packaged together. It is also known as _____.

(05 marks)

2. a) What is an algorithm? (02 marks)

b) Briefly describe any four (04) types of network topologies in computer network, including their advantages and disadvantages. (08 marks)

c) Draw a flowchart for an algorithm to

- Allow numbers to be input
- Ignore any numbers that are negative
- Count how many numbers are positive
- Output the count of positive numbers when zero is input and end the algorithm

(10 marks)

d) Explain the changes you will make to your algorithm to also count the negative numbers.

(05 marks)

3. a) What happens in the computer, when you type the letter 'A' on the keyboard? Explain briefly. (Binary value for the letter 'A' is '01000001'). (04 marks)

b) Convert the following numbers from the given base to the requested base. (Show all your workings.)

- a. 162.375_{10} to binary
- b. 1101.011_2 to decimal
- c. $2E75.C2_{16}$ to octal

(06 marks)

- c) What are the five functions of the Central Processing Unit (CPU)? (05 marks)
- d) The CPU has several components, including the Memory Data Register (MDR) and the Arithmetic Logic Unit (ALU). Describe how the MDR and the ALU are used in the functions mentioned above (in 3. c.) using an example. (10 marks)
4. a) What are the functions of an Operating System? Briefly explain three (03) of them. (06 marks)
- b) Give two (02) examples of System Software. (02 marks)
- c) Explain the concept of Public Domain Software License and its permissions. (06 marks)
- d) What are the primary characteristics of a Virus, and how does it spread across computers? (05 marks)
- e) What are the different strategies that can be used to protect yourself from computer crimes? (06 marks)

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